Forecasting and Analyzing Stock Prices Using Machine Learning Models

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Motivation and Delivery

Motivation !

1- Empowering investors, managing risk, informing decisions, and enhancing efficiency.

2- The stock market is crucial for investing, retirement planning, and asset growth.

3- Understanding stock market dynamics and applying machine learning approaches will enable consumers to navigate the market.

Cont..

Delivery!

1-The model utilizes real-time data.
2- Users can stay informed about the dynamic market.
3- The model serves anyone interested in the stock market, regardless of their experience level.
4- It offers valuable information and analysis for making informed choices.
5- It helps users discover potential market opportunities and manage risks.
6- Users can optimize their investment plans with the help of the model.

Design Project

- Structure of the Project:

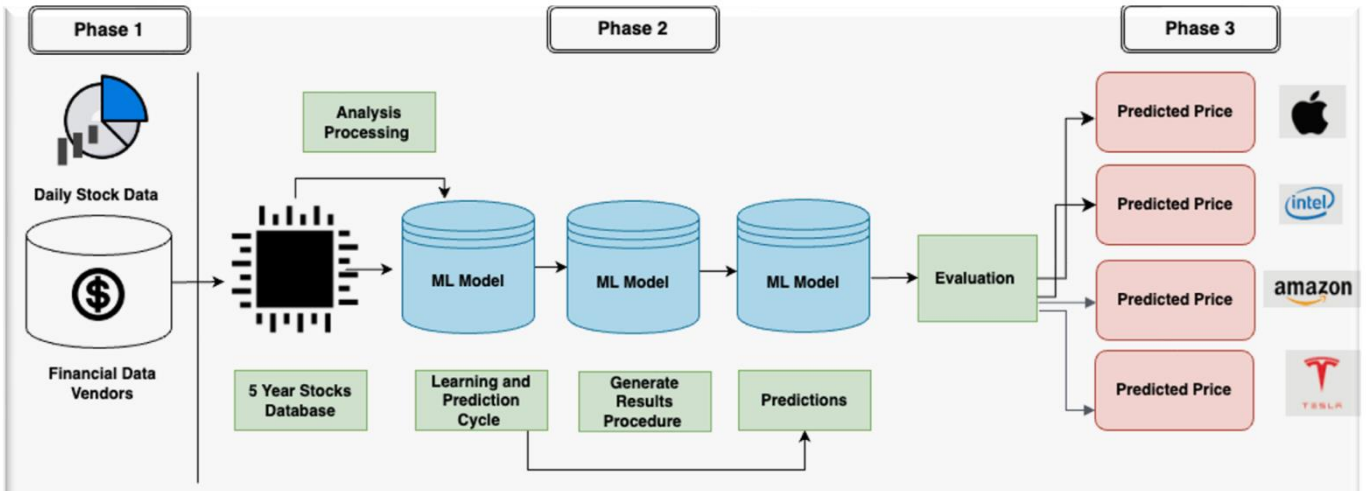
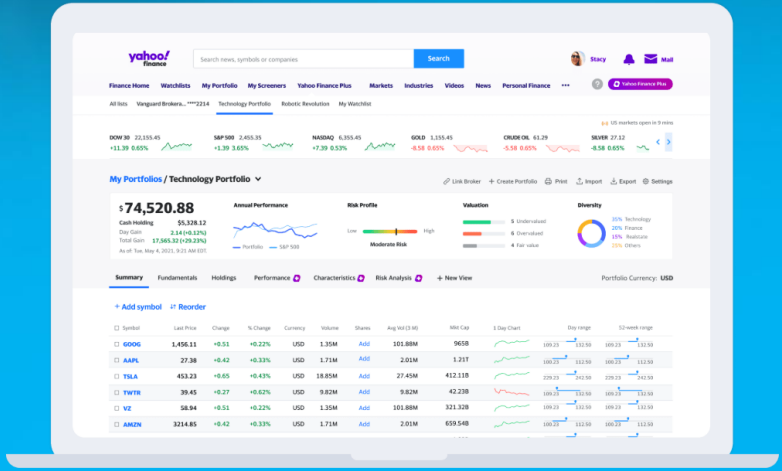


Figure 1 : Flow Diagram for Forecasting and Analyzing Stock Prices Using a Machine Learning Models

Cont..

- How was the data acquired!



- What are the features!

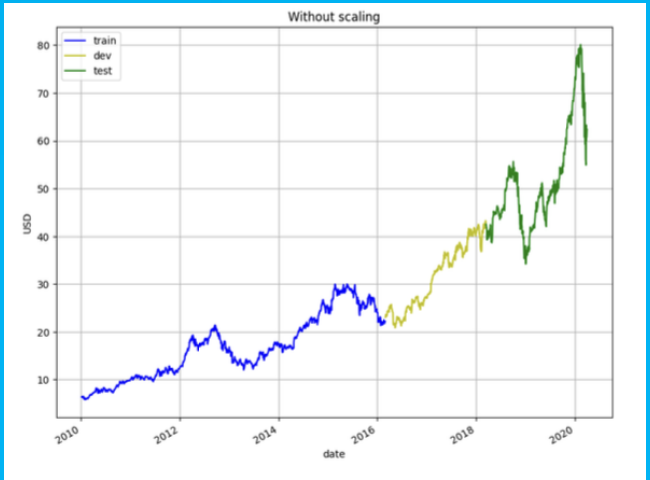
```
Index(['adj_close', 'volume', 'date', 'month', 'range_hl', 'range_oc',
       'order_day', 'adj_close_lag_1', 'range_hl_lag_1', 'range_oc_lag_1',
       'volume_lag_1', 'adj_close_lag_2', 'range_hl_lag_2', 'range_oc_lag_2',
       'volume_lag_2', 'adj_close_lag_3', 'range_hl_lag_3', 'range_oc_lag_3',
       'volume_lag_3', 'adj_close_mean', 'volume_mean', 'adj_close_std',
       'range_hl_mean', 'range_hl_std', 'range_oc_mean', 'range_oc_std',
       'volume_std'],
      dtype='object')
```

Cont..

- The numbers of observation:

Sets	Size	Shape
Training	1546	(1546, 12)
Validation	514	(514, 12)
Testing	514	(514, 12)
Training+ validation (CV)	2060	(2060, 12)

The dataset is spitted into
60% for training,
20% for validation,
and 20% for testing sets.



Cont..

- Productor Variable :

Target column is adj_close and the selected features are:

```
features: ['adj_close_lag_1', 'range_hl_lag_1', 'range_oc_lag_1', 'volume_lag_1', 'adj_close_lag_2', 'range_hl_lag_2', 'range_oc_lag_2', 'volume_lag_2', 'adj_close_lag_3', 'range_hl_lag_3', 'range_oc_lag_3', 'volume_lag_3']
```

- What models were used!

XGBoost

Random forest

- Were there any particularly important design decisions ?

Creating new features that are not provided in dataset because the model's results based on the original features were very low.

Cont..

What’s next?

Considering daily, weekly, and monthly analyzing prices. Also, creating other features to calculate the movement of prices.

Demonstration

Comparison and Evaluation

Metrics	RFR	XGBoost
MSE	8.0919	1.493
RMSE	2.8446	1.222
MAE	1.7099	0.753

Demo..